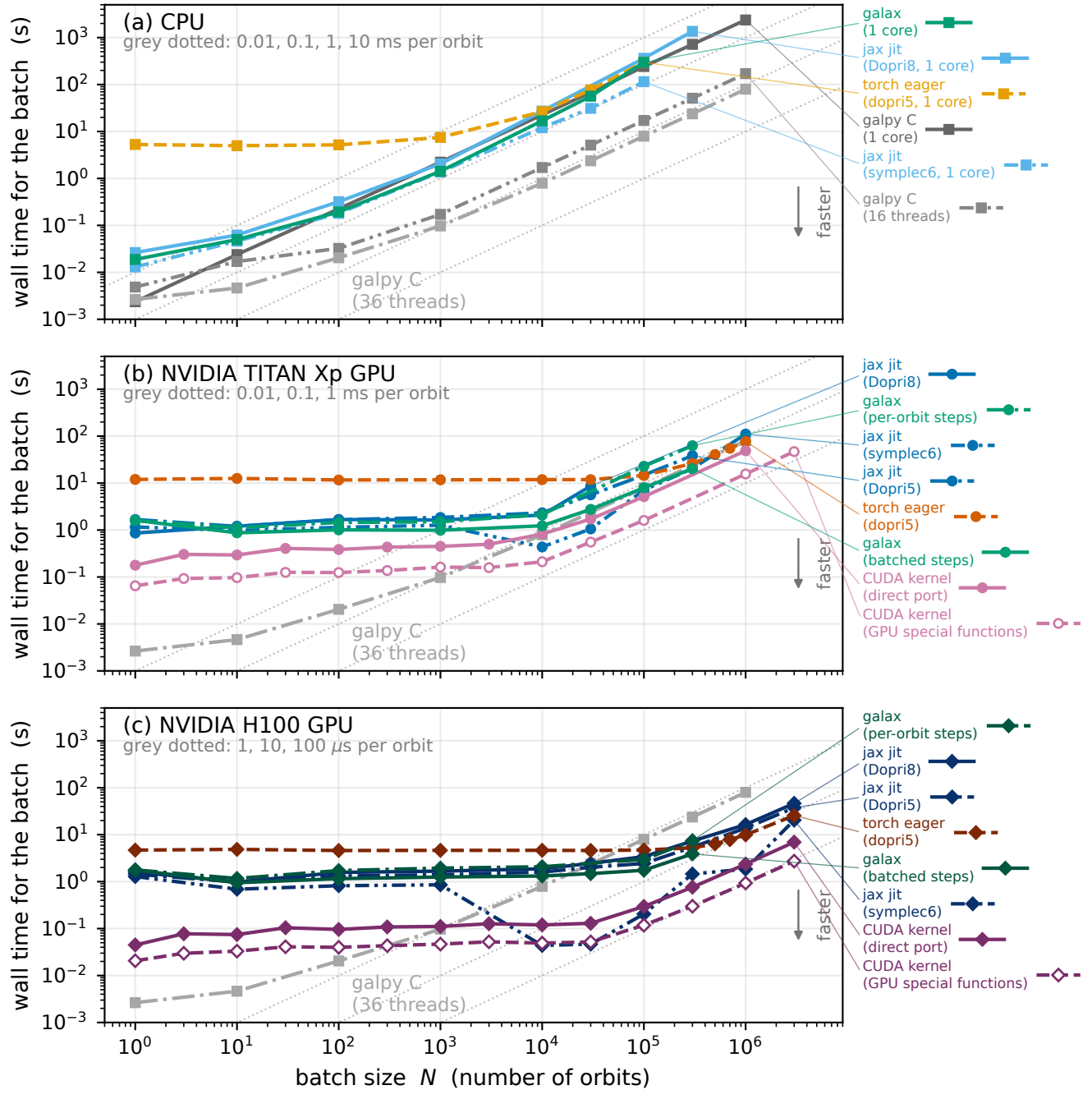
(a) CPU
grey dotted: 0.01, 0.1, 1, 10 ms per orbit
galpy C
(36 threads)
faster
galax
(1 core)
jax jit
(Dopri8, 1 core)
torch eager
(dopri5, 1 core)
galpy C
(1 core)
jax jit
(symplec6, 1 core)
galpy C
(16 threads)
(b) NVIDIA TITAN Xp GPU
grey dotted: 0.01, 0.1, 1 ms per orbit
galpy C
(36 threads)
faster
jax jit
(Dopri8)
galax
(per-orbit steps)
jax jit
(symplec6)
jax jit
(Dopri5)
torch eager
(dopri5)
galax
(batched steps)
CUDA kernel
(direct port)
CUDA kernel
(GPU special functions)
(c) NVIDIA H100 GPU
grey dotted: 1, 10, 100 μs per orbit
galpy C
(36 threads)
faster
galax
(per-orbit steps)
jax jit
(Dopri8)
jax jit
(Dopri5)
torch eager
(dopri5)
galax
(batched steps)
jax jit
(symplec6)
CUDA kernel
(direct port)
CUDA kernel
(GPU special functions)
wall time for the batch (s)
batch size N (number of orbits)